

Exercises for Process Mining

Chiara Di Francescomarino

a.y. 2025/2026

University of Trento

Davide Donà
Andrea Blushi

January 2026

Notes Repository

Contents

1	Process Discovery	1
1.1	Alpha Miner	1
1.2	Heuristic Miner	3
1.3	Region Based Miner	5
1.4	Inductive Miner	7
2	Conformance Checking	9
2.1	Replay-based Conformance Checking	9
2.2	Alignment-based Conformance Checking	12

1 Process Discovery

1.1 Alpha Miner

Exercise 1

Given the following event log:

$$L_1 = [\langle a, b, c, d \rangle^3, \langle a, c, b, d \rangle^2, \langle a, e, d \rangle^1]$$

build the process model using the Alpha Miner algorithm.

Solution

1.

Exercise 2

Given the following event log:

$$L = [\langle a, b, c, d, e, f, b, d, c, e, g \rangle, \langle a, b, d, c, e, g \rangle^2, \\ \langle a, b, c, d, e, f, b, c, d, e, f, b, d, c, e, g \rangle]$$

Use the Alpha Miner algorithm to discover a Petri net from the event log.

Solution:

1. First, we need to compute the footprint matrix:

	A	B	C	D	E	F	G
A	#	→	#	#	#	#	#
B		#	→	→	#	←	#
C			#		→	#	#
D				#	→	#	#
E					#	→	→
F						#	#
G							#

Table 1: Footprint Matrix for the Event Log L

2. Next, we identify the sets of all transitions T_L , the set of start activities T_I and the set of end activities T_O :

- $T_L = \{a, b, c, d, e, f, g\}$
- $T_I = \{a\}$

- $T_O = \{g\}$

3. Then, we identify the pairs of sets of transitions that satisfy the Alpha Miner conditions. To find the first pairs, we look for causality relations in the footprint matrix:

$$\{(\{a\}, \{b\}), (\{b\}, \{c\}), (\{b\}, \{d\}), (\{f\}, \{b\}), (\{c\}, \{e\}), (\{d\}, \{e\}), (\{e\}, \{f\}), (\{e\}, \{g\})\}$$

4. We can then group these pairs into maximal sets:

$$\{(\{a, f\}, \{b\}), (\{e\}, \{f, g\})\}$$

And we can remove the non-maximal sets that composed them:

$$\{(\{a\}, \{b\}), (\{f\}, \{b\}), (\{e\}, \{f\}), (\{e\}, \{g\})\}$$

5. Finally, we can construct the Workflow net, using one place for each of the identified pairs, and connecting them according to the causality relations:

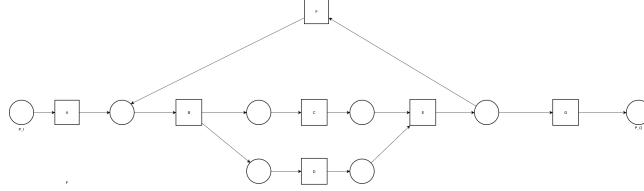


Figure 1: Petri Net Discovered using the Alpha Miner

Exercise 3

Given the following event log:

$$L_4 = [\langle a, c, d \rangle^{45}, \langle b, c, d \rangle^{42}, \langle a, c, e \rangle^{38}, \langle b, c, e \rangle^{22}]$$

build the process model using the Alpha Miner algorithm.

Solution

- 1.

Exercise 4

Consider the following event log:

$$L = [\langle a, b, d \rangle^{100}, \langle a, c, b, d \rangle^{80}, \langle a, b, c, d \rangle^2, \langle a, c, d \rangle^{40}, \langle a, d \rangle^1]$$

Use the Alpha Miner algorithm to discover a Petri net from the event log.

Solution:

1.2 Heuristic Miner

Exercise 1

Given the following event log:

$$L = [\langle a, e \rangle^5, \langle a, b, c, e \rangle^{10}, \langle a, c, b, e \rangle^{10}, \langle a, b, e \rangle^1, \langle a, c, e \rangle^1, \\ \langle a, d, e \rangle^{10}, \langle a, d, d, e \rangle^2, \langle a, d, d, d, e \rangle^1]$$

use the Heuristic Miner algorithm to discover a process model from the event log, with a threshold of 0.7.

Solution:

1. First, we need to compute the frequency matrix:

	A	B	C	D	E
A	0	11	11	13	5
B	0	0	10	0	11
C	0	10	0	0	11
D	0	0	0	4	13
E	0	0	0	0	0

Table 2: Frequency Matrix for the Event Log L

Each cell (x, y) contains the number of times activity x is directly followed by activity y in the event log.

2. Next, we compute the dependency matrix, by applying the dependency measure formula to each pair of activities. Remember that the dependency measure is defined as:

$$\begin{cases} \frac{|x \rightarrow y| - |y \rightarrow x|}{|x \rightarrow y| + |y \rightarrow x| + 1} & \text{if } x \neq y \\ \frac{|x \rightarrow x|}{|x \rightarrow x| + 1} & \text{otherwise} \end{cases}$$

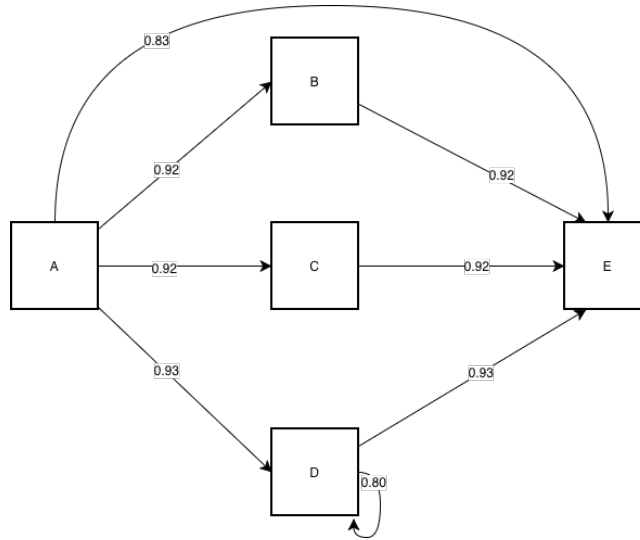
Note that:

- Along the diagonal, we apply the second case of the formula;
- We can compute the upper triangle of the matrix, the lower triangle can be derived from it by simply swapping signs.

	A	B	C	D	E
A	0	$\frac{11}{12}$	$\frac{11}{12}$	$\frac{13}{14}$	$\frac{5}{6}$
B		0	$\frac{10-10}{21}$	0	$\frac{11}{12}$
C			0	0	$\frac{11}{12}$
D				$\frac{4}{5}$	$\frac{13}{14}$
E					0

Table 3: Dependency Matrix for the Event Log L

3. We can then build the dependency graph, using the given threshold. We create a node for each activity in the log, and add the directed edges for each pair of activities whose dependency measure is above the threshold:

Figure 2: Dependency Graph for the Event Log L **Exercise 2**

Given the following event log:

$$L_2 = [\langle a, b, d, e \rangle^{80}, \langle a, d, b, e \rangle^{30}, \langle a, e \rangle^2, \langle a, b, c, d, e \rangle^{10}]$$

use the Heuristic Miner algorithm to discover the dependency matrix from the event log and build the dependency graph with a threshold of 0.9.

Exercise 3

Given the following event log:

$$L_3 = [\langle a, b, d \rangle^{100}, \langle a, c, b, d \rangle^{80}, \langle a, b, c, d \rangle^2, \langle a, c, d \rangle^{40}, \langle a, d \rangle^1]$$

Compute:

- The frequency matrix;
- The dependency matrix;
- The dependency graph with a threshold of 0.9 and a threshold for direct succession of 1.

1.3 Region Based Miner**Exercise 1**

Construct a transition system applying the *past with multiset abstraction* for the given log:

$$L = [\langle a, c, b, d \rangle, \langle a, b, c, d \rangle, \langle b, a, c, d \rangle]$$

Solution:

1. We create a starting state q_0 with an empty multiset;
2. We replay each trace in the log, updating the multiset and creating new states as needed. For instance, considering the first trace $\langle a, c, b, d \rangle$:
 - For the event a , we add a transition from q_0 to a new state q_1 with multiset $\{a\}$;
 - For the event c , we add a transition from q_1 to q_2 with multiset $\{a, c\}$;
 - For the event b , we add a transition from q_2 to q_3 with multiset $\{a, b, c\}$;
 - ...

Continuing this process for all traces, we obtain the following transition system:

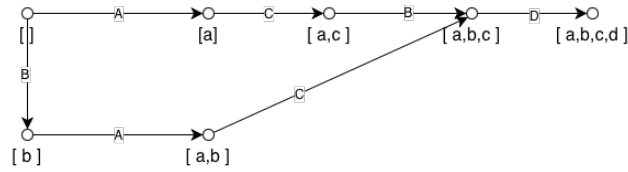


Figure 3: Transition System using Past with Multiset Abstraction

Exercise 2

Given the previous transition system, compute the resulting Petri net.

Solution:

1. We identify the minimal, non-trivial regions in the transition system. In this case, we have:

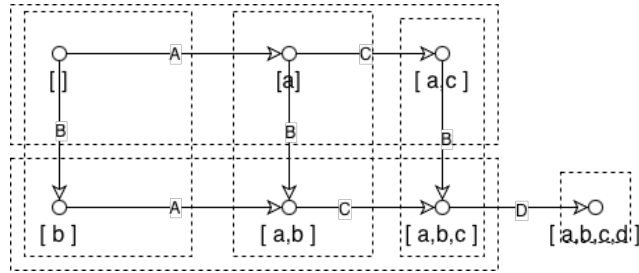


Figure 4: Identified Regions from the Transition System

2. We then create a place for each region. All the transition entering a region will have an arc to the corresponding place, and all transitions leaving the region will have an arc from the place. This results in the following Petri net:

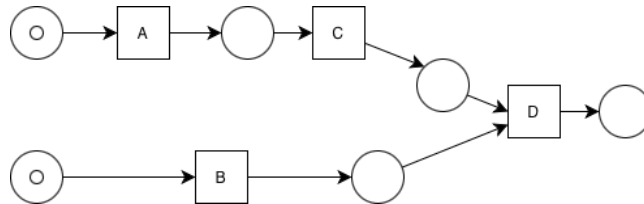


Figure 5: Petri Net Derived from the Transition System

Exercise 3

Given the following event log:

$$L = [\langle a, b, c, d \rangle^3, \langle a, c, b, d \rangle^2, \langle a, e, d \rangle^1]$$

build the process model using the Region Based Miner algorithm with *past with set abstraction*.

Exercise 4

Given the following event log:

$$L = [\langle a, b, d \rangle^{100}, \langle a, c, b, d \rangle^{80}, \langle a, b, c, d \rangle^2, \langle a, c, d \rangle^{40}, \langle a, d \rangle^1]$$

Use the Region Based Miner algorithm with *past with set abstraction* to discover a Petri Net.

Solution:

1.4 Inductive Miner**Exercise 1**

Given the following event log:

$$L = [\langle a, b, c, d \rangle^3, \langle a, c, b, d \rangle^4, \langle a, b, c, e, f, b, c, d \rangle^2, \langle a, c, b, e, f, b, c, d \rangle^2, \\ \langle a, b, c, e, f, c, b, d \rangle, \langle a, c, b, e, f, b, c, e, f, c, b, d \rangle]$$

use the Inductive Miner algorithm to discover a process model from the event log.

Exercise 2

Given the following event log:

$$L = [\langle a, c, d \rangle^{45}, \langle b, c, d \rangle^{42}, \langle a, c, e \rangle^{38}, \langle b, c, e \rangle^{22}]$$

use the Inductive Miner algorithm to discover a process model from the event log.

Solution:

Exercise 3

Given the following event log:

$$L = [\langle a, b, d \rangle^{100}, \langle a, c, b, d \rangle^{80}, \langle a, b, c, d \rangle^2, \langle a, c, d \rangle^{40}, \langle a, d \rangle^1]$$

use the Inductive Miner algorithm to discover a process model from the event log.

2 Conformance Checking

2.1 Replay-based Conformance Checking

Exercise 1

Given the following Petri net and event log:

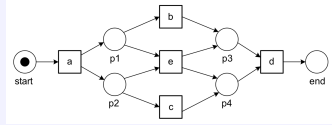


Figure 6: Petri net for Exercise 1

$$L = [\langle a, b, c, d \rangle^{10}, \langle a, c, b, d \rangle^{10}, \langle a, e, d \rangle^{10}, \langle a, b, d \rangle^2, \langle a, c, d \rangle^1, \\ \langle a, d \rangle^1, \langle a, b, b, d \rangle^1]$$

Use replay-based conformance checking to compute the fitness of the event log with respect to the Petri net.

Solution:

1. First, we replay each unique trace in the event log on the Petri net and collect the statistics:
 - For trace $\langle a, b, c, d \rangle$:
 - We first produce a token in place *start* ($p = 1$);
 - We then fire transition *a*, consuming the token from place *start* and producing a token in each of the output places of *a* ($c = 1$, $p = 3$);
 - We then fire transition *b*, consuming a token from place p_1 and producing a token in p_3 ($c = 2$, $p = 4$);
 - We then fire transition *c*, consuming a token from place p_2 and producing a token in p_4 ($c = 3$, $p = 5$);
 - We then fire transition *d*, consuming tokens from places p_3 and p_4 and producing a token in place *end* ($c = 5$, $p = 6$);
 - Finally, we consume the token from place *end* ($c = 6$, $p = 6$);
 - For trace $\langle a, c, b, d \rangle$, we have the same statistics as above: $p = 6$, $c = 6$, $m = 0$, $r = 0$;
 - For trace $\langle a, e, d \rangle$, we have $p = 6$, $c = 6$, $m = 0$, $r = 0$;
 - For trace $\langle a, b, d \rangle$, we have $p = 5$, $c = 5$, $m = 1$, $r = 1$;

- For trace $\langle a, c, d \rangle$, we have the same statistics as above: $p = 5, c = 5, m = 1, r = 1$;
 - For trace $\langle a, d \rangle$, we have $p = 4, c = 4, m = 2, r = 2$;
 - For trace $\langle a, b, b, d \rangle$, we have $p = 6, c = 6, m = 2, r = 2$;
2. Now, we can sum the statistics weighted by the frequency of each trace in the event log:
- $P_{total} = 10 \times 6 + 10 \times 6 + 10 \times 6 + 2 \times 5 + 1 \times 5 + 1 \times 4 + 1 \times 6 = 205$;
 - $C_{total} = 205$ since every trace has the same number of produced and consumed tokens;
 - $M_{total} = 0 + 0 + 0 + 1 \times 2 + 1 + 2 + 2 = 7$;
 - $R_{total} = 7$ since the remaining tokens are equal to the missing tokens for each trace.
3. Finally, we can compute the fitness of the event log using the formula:

$$\begin{aligned}
 Fitness_{Log} &= \frac{1}{2} \left(1 - \frac{M_{total}}{C_{total}}\right) + \frac{1}{2} \left(1 - \frac{R_{total}}{P_{total}}\right) = \frac{1}{2} \left(1 - \frac{7}{205}\right) + \frac{1}{2} \left(1 - \frac{7}{205}\right) = \\
 &= 1 - \frac{7}{205} = \frac{198}{205} \approx 0.9659
 \end{aligned}$$

Exercise 2

Given the following event log:

$$L = [\langle a, b, d, e, f \rangle^{10}, \langle a, c, e, d, f \rangle^{10}]$$

1. Construct the Petri net using the Alpha Miner algorithm;
2. Use replay-based conformance checking to compute the fitness of the event log with respect to the Petri net.

Exercise 3

Given the following Petri net and event log:

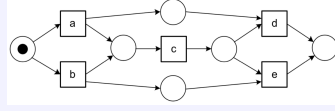


Figure 7: Petri net for Exercise 3

$$L = [\langle a, c, d \rangle^{10}, \langle b, c, e \rangle^{10}], \langle a, c, e \rangle^5, \langle b, c, d \rangle^5, \langle d, c, a \rangle, \\ \langle a, b, d \rangle, \langle d \rangle]$$

Use replay-based conformance checking to compute the fitness of the event log with respect to the Petri net.

Solution:

Exercise 4

Given the following Petri net and event log:

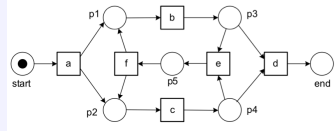


Figure 8: Petri net for Exercise 4

$$L = [\langle a, b, e, f, c, d \rangle^{10}, \langle a, b, b, e, f, c, c, d \rangle^{10}]$$

Compute the fitness of the event log with respect to the Petri net using replay-based conformance checking.

Solution:

Exercise 5

Given the following Petri net and trace:

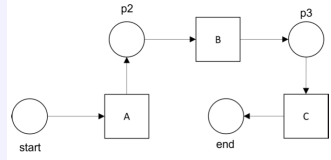


Figure 9: Petri net for Exercise 5

Trace: $\langle a, b, b \rangle$ Compute the number of produced, consumed, missing, and remaining tokens.

Solution:

Solution:

2.2 Alignment-based Conformance Checking

Exercise 1

Given the following Petri net and event log:

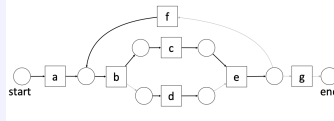


Figure 10: Petri net for Exercise 1

$$L = [\langle a, b, c, d, e, g \rangle^{10}, \langle a, b, c, e, f, e, g \rangle^2]$$

Use alignment-based conformance checking to compute the fitness of the event log with respect to the Petri net.

Solution:

1. First, we need to compute the fitness for each unique trace in the event log:
2. For trace $\langle a, b, c, d, e, g \rangle$, suppose it matches perfectly with the process model:
 - Optimal alignment cost = 0;

- To compute the worst-case alignment cost, we first need to make all log moves, resulting in a cost of 6 (one for each event in the trace);
- Then, we need to make all model moves in the shortest path from the initial to the final marking, which is 6.

a	b	c	d	e	g	»	»	»	»	»	»
»	»	»	»	»	»	a	b	d	c	e	g

- Worst-case alignment cost = 12;
 - Fitness = $1 - \frac{0}{12} = 1$;
3. For trace $\langle a, b, c, e, f, e, g \rangle$:
- Optimal alignment cost = 3, 2 log moves (for e and f) and 1 model move (for d);

a	b	c	»	e	f	e	g
a	b	c	d	»	»	e	g

- Worst-case alignment cost = 13, 7 log moves and 6 model moves;
4. Finally, we can compute the fitness of the event log as the weighted average of the fitness of each trace:

$$fitness(L, N) = \frac{10 \times 1 + 2 \times 0.75}{10 + 2} = \frac{11.5}{12} \approx 0.96$$